

Segment 5 — SpO2 Ratio-of-Ratios & Empirical Calibration

Spandan: Contactless Vital Sign Monitoring — EEE 312 DSP Project, BUET

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August 6, 2026

Why This Segment Exists

Segments 2-4 got us a clean pulse signal and a heart rate. This segment tackles a genuinely different measurement: **SpO2**, the percentage of hemoglobin in arterial blood that is carrying oxygen. Unlike heart rate (a *frequency*, read straight off an FFT peak), SpO2 is a *physiological ratio* that has no direct optical measurement — it has to be inferred from how differently oxygenated vs. deoxygenated blood absorbs two different wavelengths of light, and that inference has to be calibrated empirically against real ground-truth SpO2 readings, not derived from first principles alone.

Intuition First

Intuition first. A thermometer measures temperature directly — the mercury (or the thermistor) responds physically to heat, no fitting required. A pulse oximeter is not like that. It measures how much red and infrared light a fingertip (or, here, a patch of facial skin) absorbs as blood pulses through it, and then relies on a curve — fitted in advance, against real patients with real measured SpO2 — to convert that optical ratio into a percentage. Every commercial pulse oximeter on the market has this same empirical calibration curve baked into it at the factory. This segment builds the same kind of curve, from scratch, using this project’s own (much smaller) dataset.

Why AC/DC Per Channel Physically Encodes Oxygenation Information

Every heartbeat pushes a fresh pulse of arterial blood into the tissue being observed (a fingertip in classic pulse oximetry, a forehead/cheek patch in this project’s facial video). That blood pulse is genuinely extra volume, briefly, in the optical path — the tissue is very slightly “redder” or “less red” at the peak of each pulse than in between beats. Any single color channel’s brightness over time can be split into two parts:

- **DC (the mean level):** the steady, unchanging brightness contributed by everything that ISN’T the beat-to-beat blood volume change — skin, bone, venous blood, ambient lighting, camera sensor characteristics. This is the *overwhelming majority* of the signal’s total brightness.
- **AC (the pulsatile ripple):** the small amount by which brightness wiggles up and down, in sync with the heartbeat, on top of that DC level. This is a tiny fraction of the total signal (often well under 1%) but it’s the part that’s actually changing *because of* the blood pulse.

AC / DC — the pulsatile ripple expressed as a *fraction* of the channel’s own average brightness — is what pulse oximetry actually measures per channel, specifically because it cancels out how bright or dim that particular channel happened to be for reasons that have nothing to do with blood (skin tone, lighting, camera gain). Two channels (traditionally Red and Infrared) each get their own AC/DC, and the **ratio of those two ratios** is the final quantity, R, that gets mapped to SpO2:

$$R = (AC_Red / DC_Red) / (AC_Blue / DC_Blue)$$

Pause and Check Yourself

Pause and check yourself. Why divide by DC at all — why not just compare AC_Red to AC_Blue directly? (Hint: think about what would happen to a raw, un-normalized AC_Red value if the exact same subject were filmed in a much brighter room, with nothing about their actual blood oxygenation having changed.)

Why Blue Substitutes for Infrared

Classic pulse oximetry uses Red (~660 nm) and Infrared (~940 nm) light, specifically because oxygenated and deoxygenated hemoglobin absorb those two wavelengths very differently — that difference is *why* the ratio-of-ratios tracks oxygenation at all. An ordinary phone or webcam camera sensor has no infrared channel. This project (like a body of prior published remote-photoplethysmography research) substitutes the camera’s **Blue** channel for Infrared. This is worth being precise about: it is a **known, published approximation used in the rPPG literature** — not something invented for this project, and not a claim that Blue light and Infrared light interact with hemoglobin identically. It’s a practical substitution that published work has shown correlates usefully with true SpO2, with the explicit understanding that any calibration built on top of it has to be fit empirically against real ground truth (see the next section) rather than trusted from first-principles optics alone — precisely because Blue is a stand-in, not the real thing.

Common Beginner Mistake

Common beginner mistake. Treating “Blue substitutes for Infrared” as if it means Blue light physically behaves like Infrared light in hemoglobin. It doesn’t — the absorption physics is genuinely different. What’s being claimed is much narrower: that the *pattern* of Blue’s pulsatile response correlates usefully enough with true oxygenation, once empirically calibrated, to be worth using in a camera-only system that has no real IR sensor available. That’s an empirical claim, tested by calibration and validation, not a physics identity.

Why the R-to-SpO2 Relationship Must Be Fit Empirically

Even with real Red/Infrared hardware, the exact numerical relationship between R and true SpO2 is not something you derive from the Beer-Lambert absorption law and a calculator. It depends on details that are extremely hard to model exactly from physics alone: the precise spectral response curve of *this specific* camera sensor, the exact center wavelength and bandwidth of *this specific* Blue channel, skin thickness and scattering properties, sensor gain and exposure behavior, and more. Every real-world pulse oximeter manufacturer handles this by **empirical calibration**: recording R alongside a trusted reference SpO2 measurement (often a blood-gas analyzer) across many subjects and conditions, then fitting a curve (`spo2/calibrateSpO2.m` fits the simplest possible version of this: a straight line, $SpO2 = A - B \cdot R$) through the observed data.

Intuition First

Intuition first. This is exactly like calibrating a home bathroom scale. The scale doesn't derive your weight from the physics of its springs and Hooke's Law computed from scratch — the manufacturer calibrates it at the factory against known, trusted reference weights, and every unit afterward just applies that fitted relationship. If you skipped calibration and trusted a theoretical spring-physics formula instead, small real-world imperfections (spring wear, temperature, manufacturing tolerance) would throw the reading off in ways no amount of better physics-from-first-principles would catch — only checking against known-correct references does.

The Leakage Risk — and Why DATASET_1's 5 Subjects Make It Acute Here

Calibration leakage happens when a subject's own data is used both to *fit* a calibration curve and to *evaluate* how well that curve predicts that same subject. This inflates apparent accuracy in a way that has nothing to do with how the method will perform on a genuinely new, unseen subject — it's measuring “can this line reproduce a point it was literally drawn through,” not “can this line predict an unknown point.”

Common Beginner Mistake

Common beginner mistake. Fitting `calibrateSpO2.m` on all 5 DATASET_1 subjects at once, then checking how close its predictions are to those same 5 subjects' true SpO2, and reporting that closeness as the method's accuracy. A straight line fit through 5 points and then evaluated on those same 5 points will always look deceptively good — it's optimized to be close to exactly those points. This tells you nothing about subject 6, who the line has never seen.

This project's leave-one-out loop (`run_segment5_dataset1_calibration_batch.m`) exists specifically to avoid this: each of the 5 subjects is predicted using a calibration fit only on the *other* 4, rotating through all 5, so no subject ever contributes to its own prediction.

Why this is a bigger concern here than it would be with more data: DATASET_1 has exactly 5 subjects with facial-video SpO2 ground truth. Leave-one-out with only 5 subjects means every single fold's calibration is fit on just 4 points — barely more than the 2 points mathematically required to define a line at all. It also means the *whole dataset's* own SpO2 range is narrow (this project's 5 subjects span roughly 96-99%, confirmed in the Line-by-Line document's results table), so even a perfectly leakage-free fit is working with very little real variation to learn from. Leave-one-out here is a genuine, correct leakage-avoidance measure — but it does not, and cannot, turn 5 thin data points into a trustworthy calibration. Both things are true at once: the *methodology* (leave-one-out) is sound; the *data volume* it's applied to is not enough to draw strong conclusions from, and this document is explicit about that rather than letting a clean-sounding cross-validation procedure imply more confidence than 5 subjects can actually support.

Pause and Check Yourself

Pause and check yourself. Suppose DATASET_1 had 50 subjects instead of 5, still spanning a similarly narrow 96-99% true SpO2 range. Would leave-one-out cross-validation across those 50 subjects fully solve the “thin data” concern raised above, or would a real limitation remain? What specifically would still be missing?

What the Hoffman Sanity Check Does and Does NOT Prove

DATASET_1's narrow SpO2 range (96-99%) can confirm the *code* runs and produces plausible numbers, but it genuinely cannot test whether `ratioOfRatios.m/calibrateSpO2.m` respond correctly across a *wide* range of true oxygenation, because DATASET_1 simply doesn't contain much range to test against. The Hoffman et al. finger-camera oximetry dataset (github.com/ubicomplab/oximetry-phone-cam-data) was used as a **different, deliberately wide-range** dataset purely to fill that gap.

What it is: subjects with a finger from each hand pressed directly on a smartphone camera lens with the flash on, recorded while a controlled Varied FiO2 protocol (a clinically supervised mixture of oxygen and nitrogen) deliberately lowered their SpO2 over roughly 12-16 minutes, down to as low as the mid-60s%, then let it recover. Six subjects, confirmed directly by opening the dataset's own CSV files (not assumed from its README) — see `docs/HOFFMAN_DATA_FORMAT.md` for the full inspection notes.

What a good result here proves: that this segment's actual production code (`ratioOfRatios.m` and `calibrateSpO2.m`, called directly, not reimplemented) responds sensibly — in the physiologically correct direction, with a real, checkable relationship — when given real optical data spanning a genuinely wide oxygenation range. That's a **code- correctness** claim.

What it does NOT prove: anything about facial-video SpO2 accuracy. Finger-on-camera-with-flash is a completely different optical regime from facial rPPG — direct contact instead of distance, an active flash instead of ambient/room lighting, and a very different tissue thickness and blood volume behind the measurement. A good result on Hoffman data is evidence the *code* is sound; it is not evidence that the *facial pipeline* will hit similar numbers. Only DATASET_1 (and, later, VIPL-HR/PURE once access arrives) can speak to actual facial-video SpO2 accuracy.

Intuition First

Intuition first. Think of this the way you'd think about unit-testing a sorting algorithm on a big, deliberately varied list of random numbers before trusting it on your real, much smaller, oddly-distributed dataset. Confirming the algorithm correctly sorts a wide, varied test case tells you the *logic* works. It doesn't tell you anything about whether your real dataset's particular quirks (near-duplicate values, a narrow range, whatever they may be) will behave the same way — that still has to be checked on the real data directly.

The actual results obtained (per-subject correlations, the pooled-vs- per-subject discrepancy that was investigated rather than accepted at face value, and the final test-set error) are reported in full, with the real numbers, in `Segment5_LineByLine_Explanation.md` Part 4-5 — this document covers the theory; that one covers what was actually measured.

Common Pitfalls in This Segment

Confusing AC/DC granularity with formula correctness. The ratio-of-ratios *formula* is identical whether it's computed once per whole clip (DATASET_1) or once per short window (Hoffman). What changes is how much of the underlying recording each `mean()/std()` call sees. Getting the formula right and getting the granularity right are two separate judgment calls — this segment documents both, in `Segment5_LineByLine_Explanation.md` Part 0, because it's easy to assume “one function, one correct way to call it” when the right window length actually depends

on whether the physical quantity you’re measuring is expected to change within a single recording.

Trusting a pooled multi-subject fit’s correlation number without checking per-subject structure underneath it. As detailed in the Line-by-Line document, a pooled correlation across several subjects can look much weaker than any individual subject’s own correlation, purely because of between-subject baseline offsets — not because there’s no real relationship. Always check whether an unexpectedly weak aggregate number is hiding real per-subject signal before concluding “no relationship” or “bug.”

Treating a wide-range sanity check as a facial-video validation result. Covered at length above — worth repeating here because it’s the single easiest thing for a reader skimming results tables to get wrong.

How This Segment’s Outputs Feed Into Segment 6

Both this segment’s batch scripts save CSVs into `results/metrics/` — `segment5_dataset1_calibration.csv` (5 leave-one-out rows) and `segment5_hoffman_sanity_check.csv` (one row per held-out test window). Neither is Segment 6’s job. `validation/runLOSO.m`, `validation/computeMetrics.m`, and `validation/blandAltman.m` (all still untouched stubs, explicitly out of scope for this segment) are where the actual, statistically meaningful leave-one-subject-out validation happens — computing MAE, RMSE, Pearson correlation, and Bland-Altman agreement across every subject with usable ground truth, for both HR (`DATASET_1` + `DATASET_2`) and SpO2 (`DATASET_1` only, since it’s the only source of SpO2 ground truth this project currently has). This segment’s job was narrower and more foundational: get `ratioOfRatios.m` and `calibrateSpO2.m` correctly implemented, demonstrate they work on real data with a real leakage-avoidance strategy, and be explicit about exactly how much (or little) the thin `DATASET_1` sample can actually prove. Segment 6 is where the team finds out, with proper statistics and across the full available data, whether the method itself holds up.